

Howard Math 273, HW# 1,

Fall 2026; Instructor: Sam Hopkins; Due: Friday, September 11th

1. Let a_n be the sequence defined recursively by $a_0 = a_1 = 1$, and $a_n = a_{n-1} + 2a_{n-2}$ for $n \geq 2$. Find the real numbers b and c for which $a_n \sim c \cdot b^n$ as $n \rightarrow \infty$, i.e., for which $\lim_{n \rightarrow \infty} \frac{a_n}{c \cdot b^n} = 1$.

Hint: First establish that the generating function is $\sum_{n \geq 0} a_n x^n = \frac{1}{1 - x - 2x^2}$. Then find the partial fraction decomposition of this rational function, and extract coefficients.

2. (*Stanley, EC1, #1.5*) Show that, for any $k \geq 1$,

$$\sum_{n_1, \dots, n_k \geq 0} \min(n_1, \dots, n_k) x_1^{n_1} x_2^{n_2} \cdots x_k^{n_k} = \frac{x_1 x_2 \cdots x_k}{(1 - x_1)(1 - x_2) \cdots (1 - x_k) \cdot (1 - x_1 x_2 \cdots x_k)},$$

where $\min(n_1, \dots, n_k)$ means the minimum of the integers $n_1, \dots, n_k$.

3. (*Stanley, EC1, #1.66*) Let $p_k(n)$ denote the number of partitions of n into exactly k parts. Give a **bijective** proof that

$$p_0(n) + p_1(n) + p_2(n) + \cdots + p_k(n) = p_k(n + k)$$

Hint: Think about Young diagrams.

4. (*Stanley, EC1, #1.26*) Let $\bar{c}(n, m)$ denote the number of compositions of n into parts of size at most m . Show that, for any $m \geq 1$,

$$\sum_{n \geq 0} \bar{c}(n, m) x^n = \frac{1 - x}{1 - 2x + x^{m+1}}.$$